Prop-Firm "Loop" Strategy: Rules, Method, Results and Replication Guide

Research notes compiled November 2026. Repository: **EnderWiggin2027/Quantconnect-Data**, branch **claude/festive-dirac-jgT5l**. Everything here is a **simulation** of futures prop-firm accounts; numbers are model outputs, not guarantees.

1. Bottom line

- The trading strategy (an intraday ES "momentum loop") has **no edge of its own**: on real ES 2022-2026 its gross result is about 0 points per trade, and a normal-account QuantConnect backtest lost money after costs. **All modeled profit comes from the prop-firm payoff structure**: losses are capped at the evaluation fee, while payouts are real cash.
- The strategy is shaped for that structure: many small winning days, occasional larger losing days, fast passes, and payouts taken before accounts bust.
- **Best accounts found** (all 25K, end-of-day drawdown, no daily loss limit): Lucid 25K Pro (loss limit OFF), FundedNext 25K Pro, MyFundedFutures 25K.
- **Recommended plan at ~1 tick slippage** (real ES, $12,000 start): 5 x Lucid 25K Pro + 5 x FundedNext 25K Pro from day 1, add 5 x MyFundedFutures 25K when cash reaches $30,000. Median result +$16.8k at 2 months (6.3% chance of being down), +$78k at 6 months, +$176k at 12 months.
- **At 2-3 ticks slippage**: start with Lucid 25K Pro only, add FundedNext 25K Pro and MFF at $30,000 cash. Median +$6.9k / +$29k / +$94k at 2 / 6 / 12 months.
- Spreading accounts across **uncorrelated markets** (crude, gold, copper, nat gas, ES...) sharply cuts early risk in synthetic tests; this still needs real multi-market data.
- Biggest real-world unknowns: **your actual slippage**, the exact firm rules marked "assumed" below, and whether firms change their rules.

2. Core idea (why this can make money with a zero-edge strategy)

A prop evaluation is a **convex payoff**: if you fail you lose only the fee (here $85-$100); if you pass, the funded account can pay out thousands. With a random-walk strategy the chance of passing depends mostly on the **shape** of the daily P&L (win size vs loss size vs drawdown vs target), not on prediction skill. In tests, a **random-direction** version of the loop passed evaluations as often as the momentum version; the momentum signal mattered only in the funded phase (it roughly doubled payouts). The model treats firms as a repeated business: buy evaluations, pass some, extract payouts before funded accounts bust, recycle the slot.

3. The strategy - exact mechanical rules

Instrument: ES / MES futures (1 ES point = $50 per mini, $5 per micro). Regular session 09:30-16:00 ET. All times below are 1-minute bar **end** times, ET.

3.1 Daily direction (shared by every account - no hedging)

- At 10:00 ET compare the price at 10:00 (close of the 10:00 bar) with the 09:30 opening price (open of the first bar).
- Above the open: every account may only go **long** today. Below: every account may only go **short**. Equal: no trading today.
- Because all accounts trade the same direction, no account is ever opposite another (firms prohibit hedging across accounts).

3.2 The loop (per account)

- From the account's start time, enter at market (bar close) in the day's direction with a stop S points away and a target T points away.
- When the trade exits (stop or target), re-enter on the next bar in the same direction with the same S and T.
- Stop for the day when the day's closed result reaches **+cap points** (per contract, gross) or after **8 trades**.
- No new entries after 15:30 ET or within 30 minutes of an early close. Flatten everything at 15:59 (or 2 minutes before an early close).
- Simulation fill rules: entry at the signal bar close; exits checked from the next bar; if a bar touches both stop and target, the stop is assumed hit first; if a bar opens beyond the stop or target, the fill is at the bar open.

3.3 Presets ("Portfolio B" slots)

Slot	Evaluation preset: start / stop / target / daily cap	Funded preset: start / stop / target / daily cap
1	10:00 / 8 / 8 / +6 pts	10:00 / 12 / 8 / +6 pts
2	11:00 / 8 / 8 / +6 pts	11:00 / 12 / 8 / +6 pts
3	10:00 / 12 / 12 / +10 pts	10:00 / 8 / 8 / +6 pts
4	11:00 / 12 / 12 / +6 pts	10:00 / 12 / 12 / +6 pts
5	13:00 / 8 / 8 / +6 pts	10:00 / 8 / 20 / +6 pts

Each firm runs 5 accounts = slots 1-5. An account uses its slot's evaluation preset until it passes, then its slot's funded preset.

3.4 Position sizes

Account type	Evaluation size (minis) slots 1-5	Funded size (minis) slots 1-5	After first payout
25K accounts, 2-mini cap (Lucid 25K Pro, FundedNext 25K Pro, MFF 25K)	2 / 2 / 1.5 / 1.5 / 2 (= 20/20/15/15/20 micros)	1.8 / 1.8 / 2 / 1.8 / 2	Halve the funded size
Lucid 50K Flex (comparison only)	3.5 / 3.5 / 2.5 / 2.5 / 3.5	3 / 3 / 4 / 3 / 4	Halve
FundedNext 50K Flex (alternative)	3 / 3 / 2.1 / 2.1 / 3	2.6 / 2.6 / 3 / 2.6 / 3	Halve

25K sizes are Portfolio B x 0.6 (the most the 2-mini cap allows); they were chosen on synthetic data and validated on real ES. **Halving size after an account's first payout** was the one optimisation that improved both profit (+10-13%) and risk.

3.5 Account management

- Keep 5 accounts per firm running. When an evaluation busts, buy a new one the next day; when a funded account busts or closes, buy a new evaluation.

- Request every payout as soon as the firm's rules allow. Payouts go back into the shared bankroll and pay for new evaluations.
- Add the next firm only when total cash (start + payouts - fees) reaches the trigger (e.g. $30,000).
- Never let accounts hold opposite positions; skip tier-1 news on MFF funded accounts (rule).

4. Prop-firm rules as modeled

"(A)" marks an **assumption** where the screenshots did not state the rule. Re-check every rule against the firm's current terms before trading; if a rule differs, change it in scripts/prop/prop_sim.py and rerun.

Rule	Lucid 25K Pro	FundedNext 25K Pro	MyFundedFutures 25K
Price	$100 incl. tax (daily loss limit OFF)	$85 (reset $84.99)	$98 incl. tax
Eval target / drawdown	$1,250 / $1,000 EOD trailing	$1,500 / $1,000 EOD	$1,500 / $1,000 EOD
Eval daily loss limit	Optional $600 (pauses). Use OFF.	None	None
Eval consistency	None	None	30%: target = max($1,500, best day / 0.3) (A: target bump, not fail)
Min eval days	-	-	4 trading days
Max size	2 minis / 20 micros	2 minis / 20 micros	2 minis / 20 micros
Funded drawdown	$1,000 EOD, locks at +$100 once up $1,100 (A)	$1,000 EOD, locks at +$100 (A)	$1,000 EOD, locks at +$100
Payout conditions	>= $250 profit since last payout; best day <= 40% of cycle profit; $1,100 buffer not withdrawable; any day (A)	Every 3 trading days (A); best day <= 40% of cycle profit; cycle profit > 0	Daily; >= $250 net since last payout; $1,100 buffer
Payout amount	profit - $1,100, cap $1,000 (1st), $1,500 (later); min $500	50% of profit (A), cap $800, min $250 (A)	profit - $1,100, no cap (A), min $500
Split	90%	90%	90%
Account limit	5 Lucid accounts total (Pro + Flex combined)	5 (A)	5 (A)
Other	Flat by 4:45pm; no hedging; micro-scalp rule	Max withdrawal $800	No tier-1 news trading when funded

4.1 Accounts tested and rejected (or weaker)

Account	Key rules modeled	Result (real ES, 5 accounts, $12k, 1 tick)
Lucid 50K Flex ($150)	$3,000 target, $2,000 EOD, 50% consistency read as $1,500/day cap; payouts 5 days of $150+, from $1,000 profit, 50% capped $2,000; floor +$100 after 1st payout; $8k live review ($900 cashout)	Works: 12-mo median about +$84-94k but 2-mo loss chance about 18% (vs 5% for 25K Pro)
Lucid 25K Pro, loss limit ON ($84)	$600 daily loss limit pauses trading; on-demand payouts	Loses: 12-mo median +$7.7k, 24% chance of loss. The limit stops the loop's recovery trades.
Lucid 150K Flex ($250-450, A)	$9,000 target, $4,500 EOD, 40% consistency ($3,600/day), 10 minis	Worse than 50K: harder target/drawdown ratio; high loss risk unless fee about $250 and bankroll $24k
Apex 50K EOD (fee A $150)	$1,000 daily loss limit; payout needs $2,600+ (safety net), 50% consistency, $200 days, 6-payout cap	Loses at every size tested; positive only with deep fee discounts
FundedNext 50K Flex ($140, reset $85)	$2,500 target, $1,500 EOD, 40% rule bumps target; payouts 5 days $200+, 50% cap $1,500, 5-payout cap, 95% split	Works (12-mo about +$55k alone); weaker than FN 25K Pro as a partner when started on day 1
Tradeify 25K v1 ($59)	Always-on $600 loss limit, 1 mini, 35% funded consistency	Loses
Tradeify 25K v2 ($76)	No eval loss limit, 1 mini; Daily path ($600 cap, $500 limit)	Weak (+$17k/yr alone). Adds about +$11k/yr as a 4th firm - excluded by choice

5. Method (how the numbers were produced)

5.1 Data

- **Real ES:** data/ES_1m_RTH/ES_1m_RTH_2022-2026.csv.gz - continuous front-month ES (open-interest roll, raw prices on the 0.25 grid), 1-minute bars 09:31-16:00 ET (bar end time), Jan 2022 - Nov 2026, 1,249 days. Downloaded from QuantConnect with ES_download.ipynb.
- **Synthetic ES:** scripts/synth_es.py - Merton jump-diffusion + GARCH(1,1) volatility clustering + 3-state Markov regimes (calm/normal/crisis), U-shaped intraday volatility and volume taken from real ES, bid/ask bounce, 0.25 tick rounding. Calibrated to real ES daily volatility (1.07%), overnight gaps, clustering and drift. Each seed is an independent 10-year market; used for (a) choosing sizes without touching real data and (b) multi-market tests.
- Earlier phases used SPY 1-minute data x10 as an ES proxy (SPY_1m_Cleaned Parquet.parquet); real-ES results later matched those closely.

5.2 Trade tape

scripts/prop/build_tape.py simulates all 10 presets bar-by-bar for every day and records, per preset and day: points per 1 contract before costs, number of trades, and the worst intraday point (needed for daily-loss-limit rules). The QuantConnect version of the same logic (LoopTape algorithm) matched this tape exactly on 19 of 25 sampled days and within a few ticks on the rest.

5.3 Costs

Per contract per trade: $3.50 commission round trip (E-mini) + slippage of 1 or 3 ticks x $12.50. Day $ = size x (50 x points - trades x cost). Costs scale with contract count (micros modeled as fractional minis).

5.4 Monte Carlo of the prop business

- Each simulated path draws trading days at random (with replacement) from the tape; **all accounts in a path trade the same drawn day**, so correlation between accounts is preserved.
- 10,000 paths, horizons 1, 2, 6, 12 months (21/42/126/252 trading days), shared cash bankroll starting at $12,000.
- Every rule in section 4 is applied each day: fees and resets, end-of-day trailing drawdown and locks, targets, consistency rules, minimum days, payout conditions, caps, buffers, splits, payout-count limits, live-review closures, and firm start triggers.
- Reported: **median**, **5th percentile** (bad 1-in-20 outcome) and **chance of a net loss** of cash at each horizon.

5.5 Avoiding overfitting

- Strategy shapes were first selected on SPY 2018-2022 and checked on 2023-2025; sizes for each new firm were selected on **synthetic** data and only then run on real ES.
- Several searches (1,500 random portfolios; filters; entry/exit models) were checked out of sample and rejected when the improvement did not hold.
- Remaining weaknesses: real ES 2022-2026 is a single, mostly rising period; synthetic data has thinner tails than real data; firm rules are partly assumed.

6. Results

6.1 Final plans, real ES 2022-2026 (cells: median / 5th percentile / chance of loss)

Slippage	Plan	1 month	2 months	6 months	12 months
1 tick	A: Lucid Pro + FN Pro day 1, add MFF at $30k	+5,915 / -4,685 / 21.9%	+16,836 / -1,099 / 6.3%	+77,978 / +25,468 / 0.2%	+175,875 / +97,882 / 0.2%
1 tick	B: Lucid Pro day 1, add FN Pro + MFF at $30k	+3,736 / -2,429 / 20.0%	+10,868 / -29 / 5.1%	+59,296 / +15,221 / 0.1%	+157,475 / +80,815 / 0.0%
1 tick	C: all three from day 1	+7,594 / -7,436 / 25.3%	+24,288 / -3,593 / 8.4%	+89,425 / +29,819 / 1.8%	+186,956 / +100,499 / 1.7%
3 ticks	A	+2,934 / -5,925 / 33.6%	+10,607 / -5,117 / 15.3%	+46,561 / +7,737 / 2.1%	+110,847 / +40,672 / 1.2%
3 ticks	B	+2,058 / -3,100 / 30.1%	+6,875 / -2,296 / 12.4%	+28,926 / +7,263 / 0.8%	+93,843 / +29,498 / 0.1%
3 ticks	C	+3,052 / -9,154 / 37.8%	+13,841 / -10,257 / 19.1%	+55,948 / -11,930 / 6.6%	+119,364 / -11,932 / 5.9%

Highlighted rows = recommended plan for that slippage level. Reproduce with: python scripts/prop/run_final_plan.py

6.2 Single firms, real ES (5 accounts, $12k start)

Account (fee)	1 tick: 2 mo	1 tick: 6 mo	1 tick: 12 mo	3 ticks: 2 mo	3 ticks: 12 mo
Lucid 25K Pro ($100)	+10,872 / 5.1%	+39,065 / 0.0%	+81,462 / 0.0%	+6,875 / 12.4%	+55,631 / 0.0%
FundedNext 25K Pro ($85)	+6,555 / 8.6%	+23,050 / 0.2%	+47,262 / 0.0%	+3,715 / 20.2%	+31,992 / 0.3%
MyFundedFutures 25K ($98)	+6,818 / 15.4%	+27,383 / 1.2%	+58,552 / 0.1%	+3,115 / 30.0%	+33,110 / 1.6%
Lucid 50K Flex ($150)	+10,132 / 17.9%	+43,986 / 1.6%	+93,582 / 0.6%	+4,775 / 31.1%	+57,137 / 4.0%

Cells: median / chance of loss.

6.3 Spreading accounts across independent markets (synthetic, upper bound; Plan C = all three firms from day 1)

Slippage	Markets	1 month	2 months	6 months	12 months
1 tick	1	+10,916 / -5,356 / 16.2%	+30,648 / +2,276 / 3.4%	+107,551 / +52,279 / 0.4%	+221,970 / +142,448 / 0.4%
1 tick	5	+9,530 / -2,003 / 9.2%	+26,291 / +7,537 / 0.9%	+92,679 / +58,049 / 0.0%	+191,753 / +141,822 / 0.0%
1 tick	10	+9,829 / +722 / 3.4%	+26,799 / +11,718 / 0.1%	+93,674 / +65,594 / 0.0%	+193,286 / +152,530 / 0.0%
3 ticks	1	+5,683 / -7,704 / 28.5%	+18,671 / -5,209 / 10.8%	+70,359 / +19,562 / 2.2%	+147,500 / +72,359 / 2.0%
3 ticks	10	+5,001 / -2,677 / 15.8%	+15,933 / +2,979 / 1.8%	+59,270 / +34,723 / 0.0%	+124,160 / +88,480 / 0.0%

Synthetic markets are perfectly independent, so this overstates the benefit. Lucid lists about 5-6 roughly independent groups (stock indexes, crude, nat gas, gold/silver, platinum, copper). The loop must be re-scaled to each market's volatility and session before use. Reproduce: run_final_plan.py --multimarket

6.4 What helped and what did not

Helped	Did not help (tested, rejected)
Loop (re-enter after exit, small daily profit cap) instead of one trade per day: about 3x more money	Day filters (VIX, overnight gap, opening-range size), fewer accounts, slower evaluation buying, spending stop-loss
Halving funded size after the first payout: +10-13% profit, lower risk	Other entries/exits: pullback, breakout, trailing stop, breakeven stop, time exits; VWAP-fade and gap-fill signals

Helped	Did not help (tested, rejected)
25K accounts with easy target/drawdown ratio (1.25-1.5) and no daily loss limit	Brute-force search of 1,500 structure mixes (did not hold out of sample); martingale; sizing up after profit
Adding firms only after cash reaches about $30k (more money, same early risk)	Daily loss limits (any firm): cut the loop's recovery trades
Different start times / stop-target shapes per account (diversification within ES)	Delaying the first payout; changing base sizes away from Portfolio B
Independent markets per account (synthetic): early risk cut by 3-10x	Same-direction "day-locked" variants could not reach 1-3% risk at 2 months on one market

7. Operating checklist

- **1.** Paper-trade the loop on a free simulated account for 2+ weeks; measure real slippage on stop exits (ticks per trade). Choose Plan A (about 1 tick) or Plan B (2-3 ticks).
- **2.** Confirm every "(A)" rule in section 4 with each firm, plus account limits and whether running the same strategy on many accounts is allowed.
- **3.** Start: 5 x Lucid 25K Pro with the daily loss limit OFF (and 5 x FundedNext 25K Pro on day 1 for Plan A). Sizes per section 3.4.
- **4.** Each day: set direction at 10:00 ET; run each account's preset; flat by 15:59; no opposite positions anywhere.
- **5.** Request payouts as soon as allowed; halve that account's funded size after its first payout; rebuy busted slots the next day.
- **6.** When cash reaches $30,000 add the next firm(s). Track results against the medians in section 6; if live results fall well below the 5th percentile, stop and re-measure costs.
- **7.** Next research step: real CL, GC, SI, HG, NG 1-minute data (multi_market_download.ipynb, split by year) to test cross-market diversification.

8. Risks and limitations

- **No edge:** profit depends entirely on firm rules. Rule changes, fee changes, payout denials or bans for "farming" behaviour would remove it.
- **Correlation:** 15 accounts on one strategy, one direction, one market lose together on bad days. Multi-market spreading helps only if real correlations are low.
- **Slippage:** 3 ticks roughly halves profit and triples early loss risk versus 1 tick. Stop orders in fast markets can be worse.
- **Data:** real ES covers 2022-2026 only; bootstrap draws days independently (ignores multi-day streaks beyond the data's own clustering).
- **Model simplifications:** instant payouts and rebuys; no platform/data fees; fractional mini sizes via micros; payout timing rules partly assumed.
- **Normal account:** the same strategy loses money with real capital (QuantConnect backtest with $100k lost most of it). Never trade it with your own capital.

9. Replication guide

9.1 Files in the repository

Path	What it is
data/ES_1m_RTH/ES_1m_RTH_2022-2026.csv.gz	Real ES 1-minute RTH bars (input)
data/tapes/ES_tape_2022-2026.csv	Daily trade tape for the 10 presets (built from the bars)
scripts/prop/presets.py	Preset definitions (section 3.3)
scripts/prop/build_tape.py	Bars -> daily tape (section 5.2)
scripts/prop/prop_sim.py	Monte Carlo of all firms and rules (section 4/5.4); edit FEE and rule constants here
scripts/prop/run_final_plan.py	Reproduces tables 6.1 (and 6.3 with --multimarket)
scripts/synth_es.py	Synthetic ES generator (jump-diffusion + GARCH + regimes)
data/synthetic/ES_synth_1m_10y_seed1.csv.gz	Example synthetic 10-year market
ES_download.ipynb, multi_market_download.ipynb	QuantConnect Research notebooks to download 1-minute futures data
docs/make_prop_pdf.py	Regenerates this PDF

9.2 Commands

```
git clone https://github.com/EnderWiggin2027/Quantconnect-Data && cd Quantconnect-Data
git checkout claude/festive-dirac-jgT5l
pip install numpy pandas pyarrow reportlab
python scripts/prop/run_final_plan.py                 # table 6.1 (about 20 s)
python scripts/prop/run_final_plan.py --multimarket   # + table 6.3 (generates 10 synthetic markets, about 15 min)
python scripts/prop/build_tape.py <bars.csv.gz> <tape.csv>   # tape for any other market file
```

9.3 Prompt to give an AI assistant

```
I have a repository (EnderWiggin2027/Quantconnect-Data, branch claude/festive-dirac-jgT5l) containing a prop-firm
strategy study. Read docs/prop_firm_strategy.pdf and the code in scripts/prop/.
1. Install numpy, pandas, pyarrow. Run python scripts/prop/run_final_plan.py and confirm the medians match
   table 6.1 of the PDF within a few hundred dollars.
2. Strategy: at 10:00 ET set the day direction (price vs 09:30 open); every account trades only that direction;
   each account loops stop/target trades from its start time until +cap points or 8 trades (presets.py).
3. Firm rules are encoded in prop_sim.py (function sim, kinds pro / fnpro / mff). Sizes are in accounts().
4. To test a new firm: add a new kind with its fee, target, drawdown, lock, consistency, payout rules, caps
   and split; choose sizes on synthetic data (scripts/synth_es.py) first; then run on the real ES tape.
5. Report median, 5th percentile and P(loss) at 1, 2, 6 and 12 months, at 1 and 3 ticks of slippage.
Keep the same anti-overfitting protocol: select on synthetic or older data, judge on data not used for selection.
```

9.4 Key parameters at a glance

```
Costs per contract per trade: commission $3.50 + ticks x $12.50 (ES tick = 0.25 pt = $12.50)
Monte Carlo: 10,000 paths, rng seed 3, bankroll $12,000, horizons 21/42/126/252 trading days
Fees: Lucid 25K Pro $100 | FundedNext 25K Pro $85 | MFF 25K $98 | Lucid 50K Flex $150 | FN 50K Flex $140 (+$85 reset)
Post-first-payout size multiplier: 0.5 | Firm start trigger: cash >= $30,000 (MFF in Plan A; FN Pro + MFF in Plan B)
```